

Abstract Algebra

: Lecture 18

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Theorem 1. If $\text{char } F = 0$, then each finite extension of F is a simple extension.

Lemma 2. Let $\text{char } F = 0$, and $E = F(\alpha, \beta)$. Then there exists $\gamma \in E$ such that $E = F(\gamma)$.

证明. Let $f(x) = \text{Irr}(\alpha, F)$ and $g(x) = \text{Irr}(\beta, F)$. Let $\gamma = \alpha + c\beta$ where $c \in F$. And let $h(x) = f(\gamma - cx) \in F(\gamma)[x]$. Then $h(\beta) = f(\gamma - c\beta) = f(\alpha) = 0$. So β is a common root of $h(x)$ and $g(x)$. Assume β is the only common root of $h(x)$ and $g(x)$. Then $x - \beta = \gcd(h(x), g(x))$ as irreducible polynomial over $\text{char} = 0$ field is separable. Hence there exists $s(x)$ and $t(x)$ s.t. $x - \beta = s(x)h(x) + t(x)g(x) \in F(\gamma)[x]$. So $\beta \in F(\gamma)$ and $\alpha = \gamma - c\beta \in F(\gamma)$, i.e. $F(\alpha, \beta) \subseteq F(\gamma)$. Conversely, $\gamma \in F(\alpha, \beta)$, so $F(\alpha, \beta) = F(\gamma)$. Suppose β' is another common root of $h(x)$ and $g(x)$. Then $0 = h(\beta') = f(\gamma - c\beta') = f(\alpha + c\beta - c\beta')$ hence $\alpha' := \alpha + c\beta - c\beta'$ is a root of $f(x)$. Thus $c = \frac{\alpha' - \alpha}{\beta - \beta'} \in F$. Take $c \in F$ which is not of the form $\frac{\alpha' - \alpha}{\beta - \beta'}$ (this is due to $|F|$ is infinity). Suppose $\deg f = m$, $\deg g = n$ then $\#\alpha' = m$, $\#\beta' = n$, $\Rightarrow \#c = \frac{\alpha' - \alpha}{\beta - \beta'} < \infty$.

证明. Let E be a finite extension of F . Then $E = F(\gamma_1, \dots, \gamma_n)$. Then $E = F(\gamma_1, \dots, \gamma_n) = F(\gamma_1)(\gamma_2, \dots, \gamma_n) = E(\gamma_2, \dots, \gamma_n) = E(\beta) = F(\gamma_1, \beta) = F(\alpha)$ is a simple extension of F for some $\alpha \in E$. $\square$

Example 3. $\mathbb{Q}(\sqrt[3]{2}, \omega)$, where $\omega = \frac{-1 + \sqrt{-3}}{2}$. $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = 6$. $\mathbb{Q}(\sqrt[3]{2}, \omega) = \mathbb{Q}(\sqrt[3]{2} + \omega)$.

Proposition 4. If F is a finite field, then each finite extension is a simple extension.

证明. Let E be a finite field extension of $F = \mathbb{F}_{p^d}$ with p prime. Then $E = \mathbb{F}_{p^n}$ with $d \mid n$, and $E^* = \langle \alpha \rangle$. So $E = F(\alpha)$. $\square$

Example 5. Let $R = \mathbb{F}_p[t]$ and let F be the fraction field of R denoted by $\mathbb{F}_p(t)$. Then $\text{char } F = p$ and $|F| = \infty$. Let $f(x) = x^p - t \in F[x]$. Suppose α is a root of $f(x)$. Then $f(x) = (x - \alpha)^p$.

Claim: $f(x)$ is irreducible in $F[x]$.

Suppose $f(x) = g(x)h(x)$, $1 \leq \deg g < \deg f$. Then $g(x) \mid f(x) = (x - \alpha)^p$, and $g(x) = (x - \alpha)^m = Ax - \alpha^m$. Thus $\alpha^m \in F$. Since $F = \mathbb{F}_p(t)$. Now $t = \alpha^p$ since $\gcd(p, m) = 1$ we have $\alpha \in F$.

Contradiction.

if $\alpha \in F \Rightarrow \alpha = \frac{p(t)}{q(t)} \in F$

$\frac{f(t)}{g(t)}$

$\deg \text{LHS} \notin \mathbb{Z}$. $\deg \text{RHS} \in \mathbb{Z}$. ∇ .

c 的选取可使得 β 为唯一公共根.

$\rightarrow$ unknown. $C \neq 0$

选取这有限个 c 之外的非 0 元素 (F) 即可令 β 是 唯一公共根 of $h(x)$ and $g(x)$.

$A \in F(x)$.

① ②

Definition 6. If all roots of f lies in E and E is the smallest extension of F , then its called the splitting field of f .

Let $f(x)$ be irreducible in $F[x]$, $\text{char } F = 0$. Let α, β be two roots of $f(x)$. Then $F(\alpha), F(\beta)$ are isomorphic field. $\alpha, \dots, \alpha_n$ are all roots of $f(x)$.

Let E be the splitting field then $E = F(\alpha_1, \dots, \alpha_n)$. And $F(\alpha_i) \simeq F(\alpha_j)$.

We consider $\text{Aut}_F(E)$ is transitive on $\{\alpha_1, \dots, \alpha_n\}$.

requirement: $f(x)$ irreducible.